0190. Reverse Bits

Easy | Category: Bit Manipulation | LeetCode: https://leetcode.com/problems/rev

Problem

- Reverse the 32 bits of an unsigned integer.

Intuition

- Read bits from the right end of n and push them onto the right end of the result, 32 times; the first bit read ends up leftmost.

Brute Force

- Convert to a binary string, pad, reverse and parse: works, but slow and clunky.

Optimal Approach

- Res = 0.
- Repeat 32 times: res = (res << 1) | (n & 1); n >>>= 1.

Worked Example

```
8-bit illustration: 00001101.

take 1 -> 1
take 0 -> 10
take 1 -> 101
take 1 -> 1011
take 0 x4 -> 10110000
```

Java Solution

```java
public class ReverseBits {
    public int reverseBits(int n) {
        int res = 0;
        for (int i = 0; i < 32; i++) {
            res = (res << 1) | (n & 1);
            n >>>= 1;
        }
        return res;
    }
}
```

Complexity

- Time: O(32). Space: O(1).

Pitfalls

- Use the unsigned shift >>> so the sign bit does not smear.